Fiscal Policy with Heterogeneous Agents II: Taxation and Labor Supply

Pedro Silos

Temple University

Fall 2026

From Positive to Normative

- Previously we asked what fiscal policy *does*: debt, crowding out, transitions, incidence. The machinery was positive.
- This topic asks what fiscal policy *should do*: how should the tax system look, given everything this course believes—uninsurable risk, borrowing constraints, wealth-dependent labor supply, life cycles?
- Three battles, three papers: taxes and **hours** (Prescott 2004), taxing **capital** (Conesa–Kitao–Krueger 2009), and tax **progressivity** (Heathcote–Storesletten–Violante 2017).
- Each is a confrontation between a clean benchmark and the frictions we have spent a semester building.

Plan of the Topic

1. **Prescott (2004)**: can tax wedges explain why Americans work 30% more than Europeans? The aggregate elasticity debate, replayed as policy.
2. **Conesa–Kitao–Krueger (2009)**: optimal capital taxation once life cycles, idiosyncratic risk, and borrowing constraints are real—"taxing capital? Not a bad idea after all."
3. **HSV (2017)**: how progressive should the labor tax be? An analytical framework for the insurance-vs-incentives trade-off—the model, its closed-form allocations, and the welfare decomposition.

The Fact

- In the early 2000s, market hours per working-age person were about **30% lower** in France, Germany, and Italy than in the United States.
- In the *1970s*, continental Europeans worked **as much as or more than** Americans. The gap is not ancient culture; it opened within a generation.
- Over the same period, European effective tax wedges on labor rose sharply; the U.S. wedge did not.
- Prescott's question: can the tax wedge alone account for the gap—with a *standard* growth model and no preference differences across countries?

The Model and the Wedge

- A representative household per country, preferences $\log c + \alpha \log(1 - h)$; Cobb–Douglas technology with capital share θ; no frictions, no heterogeneity—deliberately the plainest possible machine.
- The **effective marginal wedge** combines labor and consumption taxes (both tax the same margin: an extra hour buys consumption):

$$\tau = \frac{\tau^h + \tau^c}{1 + \tau^c}.$$

- Prescott's measurements: $\tau \approx 0.40$ for the U.S., $\tau \approx 0.60$ for France and Germany.
- Revenue is rebated lump-sum—a crucial modeling choice we will interrogate.

The Formula

Intratemporal condition $\alpha c/(1-h) = (1-\tau)w$ plus $w = (1-\theta)y/h$ give

$$h \;=\; \frac{(1-\tau)(1-\theta)}{(1-\tau)(1-\theta) + \alpha\,(c/y)}.$$

- Hours depend on the tax wedge and the consumption–output ratio; α is calibrated once, to U.S. hours.
- Note what does *not* appear: no country-specific preferences, no institutions, no unemployment. The formula is a one-parameter machine fed with measured wedges.

The Arithmetic

At $\theta = 0.32$, $\alpha = 1.54$, $c/y = 0.75$:

	wedge τ	**hours share h**	**relative to U.S.**
United States	0.40	0.260	1.00
Germany	0.59	0.194	0.75
France	0.60	0.190	0.73

- The measured wedge difference alone generates a **27% hours gap**—essentially the whole observed gap.
- Prescott's conclusion: no appeal to culture or institutions needed; Europeans respond to European prices.

What Makes It Work

- The formula embeds a large *long-run* labor supply response: locally, $d \ln h / d \ln(1-\tau) \approx 0.74$ at the U.S. point, and the relevant experiment (permanent wedge, revenue rebated) makes it effectively a **compensated** elasticity—the rebate hands the income effect back.
- That is why log utility, whose *uncompensated* wage elasticity is zero, still delivers a huge tax response: **taxes-with-rebates move the substitution margin only.**
- So the entire result turns on whether the aggregate compensated elasticity is of this size.

Critique I: Which Elasticity, Again

- Micro panel estimates of compensated hours elasticities are far below what the formula uses—the micro–macro gap, policy edition.
- The familiar defense is: the relevant margin is **extensive** (employment, retirement ages, weeks worked), and aggregate elasticities built from threshold crossings are much larger than intensive micro estimates (Rogerson; Chang–Kim). Europe's hours gap is indeed mostly bodies and weeks, not hours per week—retirement, youth employment, vacation weeks.
- But then the *size* of the response depends on the distribution of households near thresholds.

Critique II: What the Revenue Buys

- The lump-sum rebate is doing silent work: it makes the experiment near-compensated. If revenue instead buys **transfers conditioned on not working** (early retirement schemes, generous UI, disability), the distortion is amplified; if it buys subsidies to complements of work (childcare), it is dampened.
- Ljungqvist and Sargent's amendment: Europe's problem is not the tax alone but the tax *plus* the welfare state's non-employment benefits—the same wedge accounting, with the benefit side made explicit.
- Alesina et al.: unions and regulated hours did some of the work; Ohanian–Raffo–Rogerson, re-measuring across decades, put taxes back near the center.
- The modern reading: the wedge story is quantitatively serious, its single-elasticity implementation is not.

The Benchmark and the Question

- **Chamley (1986), Judd (1985)**: representative agent, complete markets, infinite horizon, flat taxes, commitment $\Rightarrow$ the long-run optimal capital income tax is **zero**. A constant capital tax is a wedge on future vs. current consumption that *compounds* with the horizon; the planner prefers the flat labor wedge.
- For two decades *the* policy benchmark. But its assumptions—no heterogeneity, no uninsurable risk, no life cycle, no borrowing constraints—are exactly what this course argues are first-order.
- Conesa, Kitao, Krueger (2009, *AER*): re-solve the Ramsey problem (welfare-maximizing taxes within a parametric class) in an economy with all three ingredients. The U.S. taxes capital heavily—is the data wrong, or the benchmark?

Environment and Policy Space

- **OLG with life cycle**: age-varying productivity, retirement, survival risk.
- **Risk**: idiosyncratic productivity shocks plus permanent ability types; one riskless asset; borrowing constraint at zero. Intensive-margin labor supply.
- **Government**: finances given spending with a flat capital income tax τ^k and a flexible parametric labor schedule (nesting flat, progressive, and deduction-based shapes); budget balanced; utilitarian planner, general equilibrium recomputed at each candidate system.
- Calibrated to U.S. aggregates, age profiles of earnings and hours, and the dispersion facts studied earlier. In short: the course's full economy, pointed at a normative question.

The Headline

- Optimal capital income tax: **about 36%**—paired with a nearly **flat labor tax ($\approx 23\%$) with a sizable deduction**, i.e. mild progressivity through the exemption.
- Welfare gains from the U.S. status quo are substantial by the standards of this literature.
- The sign of Chamley–Judd is reversed; the question is *which* removed assumption does the reversing.

Two Mechanisms

- **Life cycle ⇒ capital taxes proxy for age-dependent taxes.** Productivity, hours, and the effective Frisch elasticity vary with age; a Mirrleesian planner would tax by age, but age-dependent taxes are outside the policy space. Asset holdings are hump-shaped in age, so a capital tax is an implicit tax on the middle-aged and old—the age-dependence the planner wants but cannot legislate. *Second-best logic: a restricted instrument set makes "wrong" taxes optimal as substitutes for missing ones.*
- **Risk and constraints ⇒ capital is cheaper to tax.** Precautionary over-accumulation makes the marginal unit of saving less socially than privately valuable; concentrated wealth makes capital taxation redistribution with insurance value; for constrained households the Euler margin is already inoperative.

What Did the Work, and What Undoes It

- CKK's decompositions point at the **life cycle**: without idiosyncratic risk the optimal capital tax stays large. .

Transition: From the Base to the Shape

- CKK fixed the *shape* of the labor tax crudely (flat rate plus deduction) and asked about the *base*.
- The complementary question: holding the base, how **progressive** should the labor tax be? With incomplete markets, progressivity is the government selling insurance the market will not—at the price of blunted incentives.
- HSV (2017) build a framework transparent enough to see every force in closed form.

The Question and the Tax Function

Heathcote, Storesletten and Violante (2017, *QJE*): how much progressivity should the labor income tax have?

$$T(y) \;=\; y - \lambda\, y^{1-\tau} \qquad \Longrightarrow \qquad y - T(y) = \lambda\, y^{1-\tau}.$$

- Average rate $1 - \lambda y^{-\tau}$, marginal rate $1 - \lambda(1-\tau)y^{-\tau}$, and

$$\frac{1 - T'(y)}{1 - T(y)/y} \;=\; 1 - \tau \quad \text{for all } y :$$

  τ is **the** progressivity parameter—$1 - \tau$ is the elasticity of after-tax to pre-tax income. $\tau = 0$: flat tax at rate $1 - \lambda$; $\tau = 1$: full redistribution; $\tau < 0$: regressive.

- λ scales revenue and is pinned by the budget. One parameter for "how progressive"—the object the paper optimizes.

The Function Fits, and the U.S. Number

Log after-tax income is linear in log pre-tax income: $\ln(y - T) = \ln \lambda + (1 - \tau) \ln y$. Estimated on U.S. tax-and-transfer data, the fit is remarkably good, with $\tau^{US} \approx 0.18$. At $\tau = 0.181$, $\lambda = 0.85$:

income (× mean)	**0.25**	**0.5**	**1**	**2**	**4**
average rate	−9%	4%	15%	25%	34%
marginal rate	11%	21%	30%	39%	46%

- Below 0.41× mean income, households are net transfer recipients (negative average rates): the function captures taxes *and* transfers in one object.

Environment I: Preferences and Government

A continuum of households, no capital, no borrowing/saving across the uninsurable margin (by design: transparency over completeness).

- Period utility, with valued public good G:

$$u(c, h, G) \;=\; \log c \;-\; \frac{\exp\left[(1+\sigma)\varphi\right]}{1+\sigma}\, h^{1+\sigma} \;+\; \chi \log G.$$

  $1/\sigma$ is the Frisch elasticity; φ is *preference heterogeneity* (taste for leisure, variance v_φ), which generates hours dispersion the planner should *not* insure.

- Government: $G = gY$ and a balanced budget, $\int T(y_i)\, di = G$, which pins λ given (g, τ). The planner chooses (g, τ) to maximize utilitarian welfare.

- Log consumption utility is what buys closed forms: income and substitution effects of wages on hours cancel, so hours respond only to *insured* variation and to τ.

Environment II: Skills, Wages, and Risk

- **Skill investment** (Ben-Porath's margin, made static): before working, each household chooses a skill level $s \geq 0$ at a convex utility cost $v(s;\kappa)$, isoelastic with elasticity ψ; ability κ is drawn from an exponential distribution.
- **Wages**: $\log w_{it} = \log p(s_i) + \alpha_{it} + \varepsilon_{it}$.
  - α_{it}: persistent (random-walk) component, innovations with variance v_ω—*uninsurable*;
  - ε_{it}: transitory, variance v_ε—*insurable* (households share it within "islands" via state-contingent claims).
- **Production**: CES over skill types with elasticity of substitution θ, so the skill price $p(s)$ is endogenous. In equilibrium $\log p(s) = \pi_0 + \pi_1 s$: a log-linear skill premium whose slope π_1 falls as skills become abundant.
- The insurable/uninsurable split: progressivity can only add value where private insurance stops.

Equilibrium in Closed Form

With log utility and the HSV tax function, every allocation is log-linear in the state.

- **Hours**. Average hours scale as $\bar{h} \propto (1-\tau)^{1/(1+\sigma)}$—progressivity taxes effort uniformly. Hours respond to *insurable* shocks with elasticity

$$\frac{\partial \log h}{\partial \varepsilon} = \frac{1-\tau}{\sigma+\tau},$$

  and *not at all* to uninsurable shocks (income and substitution cancel). Higher τ dampens the efficient reallocation of hours toward high-ε states.

- **Consumption**. Uninsurable shocks pass through with elasticity $1-\tau$: $\text{var}(\log c) \ni (1-\tau)^2 v_\alpha$. Progressivity is partial insurance; at $\tau^{US} = 0.18$ the tax system absorbs 18% of a permanent wage shock.

- **Skills**. Investment responds to the *after-tax* return $(1-\tau)\pi_1$ with elasticity ψ; lower

The Welfare Decomposition

Utilitarian welfare, using $C = (1-g)Y$ and $G = gY$:

$$\mathcal{W}(g,\tau) \;=\; \underbrace{\log(1-g) + \chi \log g}_{\text{public-good split}} + (1+\chi)\,\log Y(\tau) \;-\; \mathbb{E}\big[\text{disutility of } h\big] \;+\; \mathbb{E}\big[\log \tfrac{c_i}{C}\big]\,.$$

- The optimal expenditure share is $g^* = \chi/(1+\chi)$, *independent of* τ: the public-good choice and the progressivity choice separate.
- $\log Y(\tau)$ falls with τ through two channels: fewer hours ($\bar{h}$) and less skill investment. The $(1+\chi)$ weight is the subtle cost: a compressed base also finances less valued G.
- The dispersion term is $-\frac{1}{2}\,\mathrm{var}(\log c)$ under log-normality, and splits into: skill-premium dispersion $\pi_1^2\,\mathrm{var}(s)$, uninsurable risk v_α, and preference heterogeneity v_φ—each scaled by $(1-\tau)^2$. Insurable ε does not appear (it is insured) but affects hours efficiency.

The Trade-off

- **For higher** τ: compresses $(1-\tau)^2 v_\alpha$ (insurance against uninsurable risk); compresses the skill-premium and initial-condition dispersion (redistribution).
- **Against**: lowers $\bar{h}$ (intratemporal distortion); lowers skill investment and output; distorts hours responses to insurable shocks (elasticity $(1-\tau)/(\sigma+\tau)$ falls); compresses the dispersion from v_φ, which is *efficient* dispersion the planner would rather keep; and shrinks the base financing valued G.
- The optimal τ^* sets the marginal insurance value against the marginal efficiency cost in these closed-form objects. Nothing is hidden in a numerical black box; the price is no wealth, no borrowing constraints.

Calibration (Approximate Values)

parameter	target	value
τ^{US}	U.S. tax-and-transfer data	0.18
g ($\Rightarrow \chi = g/(1-g)$)	G/Y	0.19 ($\chi \approx 0.23$)
σ	Frisch elasticity of 0.5	2
θ	skill premium / elasticity of substitution	≈ 3
ψ	skill-investment elasticity	≈ 0.65
υ_ω, υ_ε	PSID variance growth, cross-sectional var	small (~ 0.01–0.02)
υ_φ	residual hours dispersion	residual

- The insurable/uninsurable split is identified from the *joint* dynamics of wages, hours, and consumption (as in Topic 4): uninsurable shocks move consumption but not hours, insurable ones move hours but not consumption.
- The skill elasticities θ and ψ are the least disciplined inputs—and the most consequential

The Headline

- At the baseline calibration, optimal progressivity is **positive but below the U.S. status quo**: $\tau^* \approx 0.08$ against $\tau^{US} \approx 0.18$.
- The welfare gain from moving to τ^* is **modest**, on the order of half a percent of lifetime consumption: the welfare function is flat near the optimum because insurance gains and efficiency losses nearly offset over a wide range.
- Read the direction, not just the number: the insurance motive alone would justify *more* progressivity than observed; what pulls τ^* down is the **skill-investment margin** and the cost of financing valued G from a compressed base.
- Implied schedule at τ^*: lower marginal rates at the top, smaller transfers at the bottom, roughly the same average rate at the mean—the government buys less insurance and sells more incentives.

Sensitivity: What Moves τ^*

change	**effect on τ^***	**channel**
exogenous skills ($\psi \to 0$)	rises above τ^{US}	no investment cost
higher υ_ω	rises	more to insure
higher υ_ε	falls	hours efficiency
G wasteful ($\chi \to 0$)	rises	no base-financing cost
higher Frisch ($\sigma \downarrow$)	falls	effort more elastic
higher υ_φ	falls	efficient dispersion taxed

- The sign of $\tau^* - \tau^{US}$ is a statement about which margins you believe—above all whether skill acquisition responds to after-tax returns, and how much earnings risk is already privately insured.
- Extensions in the paper and after: political-economy (median-voter) τ; age-dependent progressivity; adding wealth and constraints (Heathcote–Tsujiyama and later work) moves τ^* back up.